

Matrices & Vectors



Addition and scalar multiplication

- 1 Let A be the matrix with rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$, and B be the matrix with rows $\langle 2, -1 \rangle$ and $\langle 0, 5 \rangle$. Find $A + B$ and $2A$.
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Matrix multiplication

- 2 Using A (rows $\langle 1, 3 \rangle, \langle -2, 4 \rangle$) and B (rows $\langle 2, -1 \rangle, \langle 0, 5 \rangle$) from above, find AB .
 - 3 Let A have rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$, and let $\mathbf{v} = \langle 3, -1 \rangle$. Find $A\mathbf{v}$.
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Determinant and inverse

- 4 Find $\det A$ for the matrix A with rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$, and state whether A is invertible.
 - 5 Find A^{-1} for the matrix A with rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$.
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Solving systems with the inverse matrix

- 6 Solve the system $x + 3y = 13$ and $-2x + 4y = 4$ using $\mathbf{x} = A^{-1}\mathbf{b}$ (using A^{-1} from above).
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Vector magnitude and dot product

- 7 For $\mathbf{u} = \langle 4, 3 \rangle$ and $\mathbf{w} = \langle -3, 4 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.
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Angle between vectors

- 8 Find the angle between $\mathbf{p} = \langle 2, 2 \rangle$ and $\mathbf{q} = \langle 3, 0 \rangle$.
 - 9 Find the angle between $\langle 7, 1 \rangle$ and $\langle 2, -3 \rangle$, rounded to the nearest tenth of a degree.
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Multiple choice — easy

- 10 Let A be the matrix with rows $\langle -8, 4 \rangle$ and $\langle 7, 7 \rangle$, and B be the matrix with rows $\langle -2, 4 \rangle$ and $\langle -7, -3 \rangle$. Find $A + B$.
- a. $\langle -8, 8 \rangle, \langle 0, 4 \rangle$
 - b. $\langle -10, 8 \rangle, \langle 0, 2 \rangle$
 - c. $\langle -10, 8 \rangle, \langle 0, 4 \rangle$
 - d. $\langle 0, 4 \rangle, \langle -10, 8 \rangle$
- 11 Let A be the matrix with rows $\langle -4, 8 \rangle$ and $\langle -9, -2 \rangle$, and B be the matrix with rows $\langle -3, -8 \rangle$ and $\langle 5, -1 \rangle$. Find $A + B$.
- a. $\langle -7, 0 \rangle, \langle -4, -3 \rangle$
 - b. $\langle -7, 0 \rangle, \langle -4, -5 \rangle$
 - c. $\langle -5, 0 \rangle, \langle -4, -3 \rangle$
 - d. $\langle -4, -3 \rangle, \langle -7, 0 \rangle$
- 12 Let A be the matrix with rows $\langle -8, -2 \rangle$ and $\langle 8, 8 \rangle$, and B be the matrix with rows $\langle -4, -8 \rangle$ and $\langle -5, 5 \rangle$. Find $A + B$.
- a. $\langle 3, 13 \rangle, \langle -12, -10 \rangle$
 - b. $\langle -10, -10 \rangle, \langle 3, 13 \rangle$
 - c. $\langle -12, -10 \rangle, \langle 3, 13 \rangle$
 - d. $\langle -12, -10 \rangle, \langle 3, 11 \rangle$
- 13 Let A be the matrix with rows $\langle 7, 4 \rangle$ and $\langle 3, 5 \rangle$, and B be the matrix with rows $\langle 9, -5 \rangle$ and $\langle -4, 3 \rangle$. Find $A + B$.
- a. $\langle 16, -1 \rangle, \langle -1, 6 \rangle$
 - b. $\langle -1, 8 \rangle, \langle 16, -1 \rangle$
 - c. $\langle 18, -1 \rangle, \langle -1, 8 \rangle$
 - d. $\langle 16, -1 \rangle, \langle -1, 8 \rangle$

- 14 Let A be the matrix with rows $\langle 1, -3 \rangle$ and $\langle 7, 6 \rangle$, and B be the matrix with rows $\langle 6, 2 \rangle$ and $\langle -5, 1 \rangle$. Find $A + B$.
- a. $\langle 7, -1 \rangle, \langle 2, 5 \rangle$
 - b. $\langle 2, 7 \rangle, \langle 7, -1 \rangle$
 - c. $\langle 7, -1 \rangle, \langle 2, 7 \rangle$
 - d. $\langle 9, -1 \rangle, \langle 2, 7 \rangle$
- 15 Let A be the matrix with rows $\langle -2, -6 \rangle$ and $\langle 5, 7 \rangle$, and B be the matrix with rows $\langle -7, 4 \rangle$ and $\langle 5, 7 \rangle$. Find $A + B$.
- a. $\langle 10, 14 \rangle, \langle -9, -2 \rangle$
 - b. $\langle -9, -2 \rangle, \langle 10, 12 \rangle$
 - c. $\langle -9, -2 \rangle, \langle 10, 14 \rangle$
 - d. $\langle -7, -2 \rangle, \langle 10, 14 \rangle$
- 16 Let A be the matrix with rows $\langle 6, -2 \rangle$ and $\langle -8, -6 \rangle$, and B be the matrix with rows $\langle -5, -4 \rangle$ and $\langle 5, -6 \rangle$. Find $A + B$.
- a. $\langle 1, -6 \rangle, \langle -3, -12 \rangle$
 - b. $\langle 3, -6 \rangle, \langle -3, -12 \rangle$
 - c. $\langle -3, -12 \rangle, \langle 1, -6 \rangle$
 - d. $\langle 1, -6 \rangle, \langle -3, -14 \rangle$
- 17 Let A be the matrix with rows $\langle -7, -9 \rangle$ and $\langle 1, -5 \rangle$, and B be the matrix with rows $\langle 7, -7 \rangle$ and $\langle -3, 6 \rangle$. Find $A + B$.
- a. $\langle 0, -16 \rangle, \langle -2, -1 \rangle$
 - b. $\langle 2, -16 \rangle, \langle -2, 1 \rangle$
 - c. $\langle -2, 1 \rangle, \langle 0, -16 \rangle$
 - d. $\langle 0, -16 \rangle, \langle -2, 1 \rangle$

- 18 Let A be the matrix with rows $\langle -3, -6 \rangle$ and $\langle -7, 1 \rangle$, and B be the matrix with rows $\langle -5, -1 \rangle$ and $\langle -4, 3 \rangle$. Find $A + B$.
- a. $\langle -6, -7 \rangle, \langle -11, 4 \rangle$
 - b. $\langle -8, -7 \rangle, \langle -11, 4 \rangle$
 - c. $\langle -8, -7 \rangle, \langle -11, 2 \rangle$
 - d. $\langle -11, 4 \rangle, \langle -8, -7 \rangle$
- 19 Let A be the matrix with rows $\langle -4, 3 \rangle$ and $\langle -7, 1 \rangle$, and B be the matrix with rows $\langle -2, 9 \rangle$ and $\langle -5, 1 \rangle$. Find $A + B$.
- a. $\langle -4, 12 \rangle, \langle -12, 2 \rangle$
 - b. $\langle -6, 12 \rangle, \langle -12, 0 \rangle$
 - c. $\langle -12, 2 \rangle, \langle -6, 12 \rangle$
 - d. $\langle -6, 12 \rangle, \langle -12, 2 \rangle$
- 20 Let A be the matrix with rows $\langle -6, 4 \rangle$ and $\langle -7, -9 \rangle$, and B be the matrix with rows $\langle 2, 6 \rangle$ and $\langle 1, -6 \rangle$. Find $A + B$.
- a. $\langle -4, 10 \rangle, \langle -6, -15 \rangle$
 - b. $\langle -2, 10 \rangle, \langle -6, -15 \rangle$
 - c. $\langle -4, 10 \rangle, \langle -6, -17 \rangle$
 - d. $\langle -6, -15 \rangle, \langle -4, 10 \rangle$
- 21 Let A be the matrix with rows $\langle 8, -7 \rangle$ and $\langle 9, -7 \rangle$, and B be the matrix with rows $\langle 7, 9 \rangle$ and $\langle 1, 9 \rangle$. Find $A + B$.
- a. $\langle 17, 2 \rangle, \langle 10, 2 \rangle$
 - b. $\langle 10, 2 \rangle, \langle 15, 2 \rangle$
 - c. $\langle 15, 2 \rangle, \langle 10, 2 \rangle$
 - d. $\langle 15, 2 \rangle, \langle 10, 0 \rangle$

- 22 Let A be the matrix with rows $\langle -3, -1 \rangle$ and $\langle 5, 1 \rangle$, and B be the matrix with rows $\langle -4, 4 \rangle$ and $\langle 9, -7 \rangle$. Find $A + B$.
- a. $\langle -5, 3 \rangle, \langle 14, -6 \rangle$
 - b. $\langle -7, 3 \rangle, \langle 14, -6 \rangle$
 - c. $\langle -7, 3 \rangle, \langle 14, -8 \rangle$
 - d. $\langle 14, -6 \rangle, \langle -7, 3 \rangle$
- 23 Let A be the matrix with rows $\langle -6, 5 \rangle$ and $\langle -9, 1 \rangle$, and B be the matrix with rows $\langle 8, 3 \rangle$ and $\langle 7, 6 \rangle$. Find $A + B$.
- a. $\langle 4, 8 \rangle, \langle -2, 7 \rangle$
 - b. $\langle 2, 8 \rangle, \langle -2, 5 \rangle$
 - c. $\langle 2, 8 \rangle, \langle -2, 7 \rangle$
 - d. $\langle -2, 7 \rangle, \langle 2, 8 \rangle$
- 24 Let A be the matrix with rows $\langle -9, 3 \rangle$ and $\langle 9, 3 \rangle$, and B be the matrix with rows $\langle 1, 9 \rangle$ and $\langle 3, -3 \rangle$. Find $A + B$.
- a. $\langle -6, 12 \rangle, \langle 12, 0 \rangle$
 - b. $\langle -8, 12 \rangle, \langle 12, 0 \rangle$
 - c. $\langle -8, 12 \rangle, \langle 12, -2 \rangle$
 - d. $\langle 12, 0 \rangle, \langle -8, 12 \rangle$

Multiple choice — medium

- 25 Let A have rows $\langle -9, 8 \rangle$ and $\langle -9, 6 \rangle$, and let $\mathbf{v} = \langle 6, 3 \rangle$. Find $A\mathbf{v}$.
- a. $\langle -36, -30 \rangle$
 - b. $\langle -28, -36 \rangle$
 - c. $\langle -30, -36 \rangle$
 - d. $\langle -30, -38 \rangle$
- 26 Let A have rows $\langle -1, -3 \rangle$ and $\langle 9, -6 \rangle$, and let $\mathbf{v} = \langle -5, 1 \rangle$. Find $A\mathbf{v}$.
- a. $\langle 2, -51 \rangle$
 - b. $\langle -51, 2 \rangle$
 - c. $\langle 4, -51 \rangle$
 - d. $\langle 2, -53 \rangle$
- 27 Let A have rows $\langle 7, 4 \rangle$ and $\langle 1, 4 \rangle$, and let $\mathbf{v} = \langle -5, 5 \rangle$. Find $A\mathbf{v}$.
- a. $\langle -15, 13 \rangle$
 - b. $\langle 15, -15 \rangle$
 - c. $\langle -15, 15 \rangle$
 - d. $\langle -13, 15 \rangle$
- 28 Let A have rows $\langle 9, -1 \rangle$ and $\langle 4, -1 \rangle$, and let $\mathbf{v} = \langle 3, -5 \rangle$. Find $A\mathbf{v}$.
- a. $\langle 17, 32 \rangle$
 - b. $\langle 32, 15 \rangle$
 - c. $\langle 34, 17 \rangle$
 - d. $\langle 32, 17 \rangle$

- 29 Let A have rows $\langle -1, 8 \rangle$ and $\langle 2, 7 \rangle$, and let $\mathbf{v} = \langle 4, -8 \rangle$. Find $A\mathbf{v}$.
- a. $\langle -48, -68 \rangle$
 - b. $\langle -66, -48 \rangle$
 - c. $\langle -68, -50 \rangle$
 - d. $\langle -68, -48 \rangle$
- 30 Let A have rows $\langle -4, 7 \rangle$ and $\langle 6, 4 \rangle$, and let $\mathbf{v} = \langle -8, 5 \rangle$. Find $A\mathbf{v}$.
- a. $\langle 69, -28 \rangle$
 - b. $\langle 67, -28 \rangle$
 - c. $\langle -28, 67 \rangle$
 - d. $\langle 67, -30 \rangle$
- 31 Let A have rows $\langle -1, 5 \rangle$ and $\langle -2, 2 \rangle$, and let $\mathbf{v} = \langle 2, -6 \rangle$. Find $A\mathbf{v}$.
- a. $\langle -32, -18 \rangle$
 - b. $\langle -16, -32 \rangle$
 - c. $\langle -32, -16 \rangle$
 - d. $\langle -30, -16 \rangle$
- 32 Let A have rows $\langle -2, 2 \rangle$ and $\langle 8, -1 \rangle$, and let $\mathbf{v} = \langle -9, -1 \rangle$. Find $A\mathbf{v}$.
- a. $\langle 18, -71 \rangle$
 - b. $\langle -71, 16 \rangle$
 - c. $\langle 16, -73 \rangle$
 - d. $\langle 16, -71 \rangle$

- 33 Let A have rows $\langle 4, -6 \rangle$ and $\langle -7, -8 \rangle$, and let $\mathbf{v} = \langle -4, -4 \rangle$. Find $A\mathbf{v}$.
- a. $\langle 8, 60 \rangle$
 - b. $\langle 60, 8 \rangle$
 - c. $\langle 10, 60 \rangle$
 - d. $\langle 8, 58 \rangle$
- 34 Let A have rows $\langle -5, -6 \rangle$ and $\langle 9, 6 \rangle$, and let $\mathbf{v} = \langle -7, -4 \rangle$. Find $A\mathbf{v}$.
- a. $\langle 59, -87 \rangle$
 - b. $\langle -87, 59 \rangle$
 - c. $\langle 59, -89 \rangle$
 - d. $\langle 61, -87 \rangle$
- 35 Let A have rows $\langle 4, -6 \rangle$ and $\langle 8, -7 \rangle$, and let $\mathbf{v} = \langle 6, -9 \rangle$. Find $A\mathbf{v}$.
- a. $\langle 111, 78 \rangle$
 - b. $\langle 78, 111 \rangle$
 - c. $\langle 80, 111 \rangle$
 - d. $\langle 78, 109 \rangle$
- 36 Let A have rows $\langle -5, -3 \rangle$ and $\langle 1, -2 \rangle$, and let $\mathbf{v} = \langle 9, 4 \rangle$. Find $A\mathbf{v}$.
- a. $\langle -57, 1 \rangle$
 - b. $\langle -55, 1 \rangle$
 - c. $\langle 1, -57 \rangle$
 - d. $\langle -57, -1 \rangle$

37 Let A have rows $\langle -3, 4 \rangle$ and $\langle -1, -6 \rangle$, and let $\mathbf{v} = \langle 1, -4 \rangle$. Find $A\mathbf{v}$.

a. $\langle -19, 21 \rangle$

b. $\langle -17, 23 \rangle$

c. $\langle -19, 23 \rangle$

d. $\langle 23, -19 \rangle$

38 Let A have rows $\langle 4, 9 \rangle$ and $\langle -8, 6 \rangle$, and let $\mathbf{v} = \langle -7, 3 \rangle$. Find $A\mathbf{v}$.

a. $\langle 1, 74 \rangle$

b. $\langle 74, -1 \rangle$

c. $\langle -1, 72 \rangle$

d. $\langle -1, 74 \rangle$

39 Let A have rows $\langle 2, -6 \rangle$ and $\langle 3, 3 \rangle$, and let $\mathbf{v} = \langle 3, 8 \rangle$. Find $A\mathbf{v}$.

a. $\langle -42, 31 \rangle$

b. $\langle -42, 33 \rangle$

c. $\langle 33, -42 \rangle$

d. $\langle -40, 33 \rangle$

Multiple choice — hard

40 Find $\det A$ for the matrix A with rows $\langle -6, 5 \rangle$ and $\langle 7, -8 \rangle$.

a. 13

b. 15

c. 11

d. -13

- 41 Find $\det A$ for the matrix A with rows $\langle -5, -3 \rangle$ and $\langle 6, 4 \rangle$.
- a. -2
 - b. 2
 - c. -4
 - d. 0
- 42 Find $\det A$ for the matrix A with rows $\langle 7, 4 \rangle$ and $\langle 3, 5 \rangle$.
- a. -23
 - b. 21
 - c. 25
 - d. 23
- 43 Find $\det A$ for the matrix A with rows $\langle 1, 8 \rangle$ and $\langle -1, 6 \rangle$.
- a. -14
 - b. 12
 - c. 14
 - d. 16
- 44 Find $\det A$ for the matrix A with rows $\langle -3, -3 \rangle$ and $\langle -3, 5 \rangle$.
- a. -26
 - b. 24
 - c. -22
 - d. -24

45 For $\mathbf{u} = \langle -12, -35 \rangle$ and $\mathbf{w} = \langle 8, 1 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

- a. $-129, 37$
- b. $-131, 37$
- c. $-131, 39$
- d. $37, -131$

46 For $\mathbf{u} = \langle -20, -21 \rangle$ and $\mathbf{w} = \langle -8, 7 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

- a. $15, 29$
- b. $13, 31$
- c. $29, 13$
- d. $13, 29$

47 For $\mathbf{u} = \langle -20, 21 \rangle$ and $\mathbf{w} = \langle -7, -5 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

- a. $29, 35$
- b. $37, 29$
- c. $35, 29$
- d. $35, 31$

48 For $\mathbf{u} = \langle -12, 35 \rangle$ and $\mathbf{w} = \langle -7, 3 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

- a. $191, 37$
- b. $189, 37$
- c. $37, 189$
- d. $189, 39$

- 49 For $\mathbf{u} = \langle 9, -40 \rangle$ and $\mathbf{w} = \langle -3, -7 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.
- a. 41, 253
 - b. 253, 41
 - c. 253, 43
 - d. 255, 41
- 50 Find the angle between $\langle 4, -8 \rangle$ and $\langle -1, 2 \rangle$, rounded to the nearest tenth of a degree.
- a. 150.0°
 - b. 165.0°
 - c. 175.0°
 - d. 180.0°
- 51 Find the angle between $\langle -5, 5 \rangle$ and $\langle 5, 2 \rangle$, rounded to the nearest tenth of a degree.
- a. 123.2°
 - b. 113.2°
 - c. 108.2°
 - d. 118.2°
- 52 Find the angle between $\langle 4, 8 \rangle$ and $\langle 3, 9 \rangle$, rounded to the nearest tenth of a degree.
- a. 13.1°
 - b. 8.1°
 - c. 3.1°
 - d. 18.1°

- 53 Find the angle between $\langle 4, 6 \rangle$ and $\langle -6, 1 \rangle$, rounded to the nearest tenth of a degree.
- a. 124.2°
 - b. 119.2°
 - c. 114.2°
 - d. 109.2°
- 54 Find the angle between $\langle 1, -2 \rangle$ and $\langle -2, -7 \rangle$, rounded to the nearest tenth of a degree.
- a. 42.5°
 - b. 47.5°
 - c. 52.5°
 - d. 37.5°